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Project Title	**Development and Evaluation of an Evidence-Grounded Multimodal RAG System for Scientific Literature**

Development and Evaluation of an Evidence-Grounded Multimodal RAG System for Scientific Literature

Abstract

Scientific answers need traceable evidence, especially when relevant full texts require institutional access. This project evaluated a multimodal retrieval-augmented generation (RAG) system over a locally managed private corpus covering ultrasound, thermal ablation, thermometry and thermal imaging. It combined provenance-preserving PDF ingestion, lexical and dense retrieval, rank fusion, CrossEncoder reranking, Gemini generation and image summaries. The corpus contained 272 papers, 3,904 page records and 4,243 chunks; DOI metadata was present for 82.94% of chunks and 5.28% had an unknown section. The author checked 50 question–evidence labels against primary sources and online records, then used Codex for a consistency check; 32 were approved and 18 corrected. Without medical-specialist review, they remain non-expert silver labels. Hybrid Recall@5 was 0.62, compared with 0.56 for lexical and 0.52 for dense retrieval. Lexical-pool reranking produced the highest mean reciprocal rank (MRR), 0.605, at 341 ms mean local latency. A 350-word/50-overlap ablation achieved Recall@5 of 0.74 but created 6,997 chunks, 64.9% more than production 650/90. Across 30 answerable and ten evidence-insufficient questions, grounded hybrid RAG scored 1.867/2 for correctness and faithfulness, with 97.6% citation precision, correct abstention on all ten evidence-insufficient cases and a 2.0% unsupported-claim rate, down from 26.9% without evidence. On 12 figures from seven Creative Commons Attribution 4.0 articles, vision summaries did not improve source Recall@5 beyond 0.917. Citation quality was 1.917 rather than 1.833, and unsupported visual inferences were zero rather than three, but no condition difference was detected. Corpus quality, retrieval and evidence constraints should therefore be improved before generator fine-tuning. The multimodal results are exploratory and do not constitute clinical validation.

Keywords: retrieval-augmented generation; multimodal learning; scientific literature; information retrieval; evidence grounding

1 Introduction

Large language models (LLMs) can explain scientific topics fluently without providing reliable evidence. They may rely on training data, combine incompatible experiments or produce a plausible statement that cannot be traced to a publication. This is a practical problem in ultrasound thermometry, thermal ablation and thermal imaging, where methods differ in maturity, range and setting. A research answer must therefore link back to its paper, page and passage.

Access creates a second problem. University subscriptions provide full texts that public tools may not index, so a general chatbot can miss papers available to a Birmingham researcher. Uploading subscription papers to an external service can also raise copyright and governance concerns. A controlled local corpus limits bulk document transfer and preserves provenance, although selected excerpts still leave the device when an external generator is used.

Retrieval-augmented generation (RAG) separates knowledge storage from generation by supplying retrieved passages as context. Lewis et al. improved several knowledge-intensive tasks with a retrievable Wikipedia index [1]. RAG still permits parsing, retrieval and generation failures, so evaluation must examine each stage rather than one opaque answer score.

This project aims to develop and evaluate a multimodal RAG system over a locally managed private corpus for evidence-grounded question answering on institution-accessible scientific literature concerning ultrasound, thermal ablation, thermometry, high-intensity focused ultrasound (HIFU) and thermal imaging. The work has six technical objectives:

1. construct a traceable research corpus with document identifiers and bibliographic metadata;
2. extract text from scientific PDFs while preserving page and section context;
3. create retrieval-ready chunks with controlled size and overlap;
4. compare lexical, dense and hybrid retrieval, followed by CrossEncoder reranking;
5. generate English answers constrained by retrieved evidence, with numbered citations and explicit uncertainty; and
6. extend and evaluate the system for exploratory image-grounded questions without claiming clinical interpretation.

These objectives lead to four research questions (RQs). RQ1 asks how lexical, dense, hybrid and reranked retrieval trade off evidence recall, ranking quality and latency on the project corpus. RQ2 examines how chunk size and overlap affect early evidence retrieval. RQ3 asks how retrieved evidence and reranking affect answer correctness, faithfulness, completeness, citation support and abstention when compared with generation without retrieval. RQ4 tests whether a question-aware visual summary improves source retrieval and grounded answering when compared with a text-only question or basic image metadata.

The resulting artefact is a FastAPI application with a Vite and React interface, SQLite storage, SentenceTransformers embeddings, hybrid retrieval, CrossEncoder reranking, Gemini generation and a question-aware image analysis path. The contribution is an auditable implementation and component-level evaluation on a specialised corpus, not a new foundation model. Generator fine-tuning would not fix failures in corpus quality, retrieval or prompting.

2 Literature Review and State of the Art

2.1 Retrieval-augmented generation and evidence grounding

Lewis et al. reported more specific and factual output than a parametric-only baseline [1]. Their Wikipedia experiments do not directly solve scientific-document QA, where papers are long, heterogeneous and conditional. This work therefore exposes intermediate evidence rather than treating RAG as a guarantee against hallucination.

Hallucination is treated as content that conflicts with, or is unsupported by, supplied evidence. LLMs can produce and fail to recognise it [2], because unconstrained generation does not establish evidential truth [3]. Retrieval reduces this risk only when context is relevant and used faithfully [4]. The prompt therefore requires numbered citations, clearly separated visual observations and abstention; code rejects citation indices outside the supplied sources.

2.2 Sparse, dense and hybrid retrieval

BM25 provides a term-weighted lexical signal [5]. Dense Passage Retrieval outperformed BM25 on several open-domain QA datasets [6], while Sentence-BERT made independently encoded text practical for similarity search [7]. Their relative value in this specialist corpus is tested rather than assumed.

Hybrid retrieval combines these signals. RRF operates on rank positions rather than raw scores [8]. This project compares it with weighted fusion because its FTS and dense scores are not naturally calibrated.

A dual encoder permits precomputed corpus vectors but scores query and passage independently. A CrossEncoder models their interaction more closely and is therefore used to rerank only a limited candidate pool. Lexical, dense and hybrid pools are tested separately because reranking cannot recover evidence omitted by the first stage.

2.3 Scientific PDF parsing, chunking and metadata

Scientific PDF extraction depends on visual layout. VILA improves structured extraction with line or block groups [9], while cross-layout tests show degradation under publisher and layout-distribution shifts [10]. Nougat instead uses visual OCR to reconstruct scientific markup [11]. This prototype uses deterministic PyMuPDF extraction and preserves title, DOI, year, page, section and chunk ID for audit.

Short passages can lose document-level context and depend strongly on split choice [12]; simply retaining more text is not sufficient because models can miss evidence in the middle of long contexts [13]. Four sizes and overlaps are therefore evaluated rather than assuming a universal default.

This system indexes titles and sections as retrieval fields, while DOI, page and chunk identifiers form its audit trail. SciRepEval shows that scientific representations can struggle to generalise across classification, regression, ranking and search formats [14]; performance on this corpus is therefore measured directly.

2.4 RAG evaluation

End-to-end correctness cannot locate a failure. RAGAS separates context, faithfulness and answer quality [4]; ARES combines synthetic judge training with a small human-labelled set [15]. A healthcare review found no standardised RAG evaluation framework [16]. This study therefore separates retrieval, citation and repeated LLM judging over non-expert silver labels.

Recall@k records whether labelled evidence appears in the first k results; MRR emphasises its first rank, and nDCG rewards relevant sources near the top. Generation is assessed for correctness, faithfulness, completeness, abstention, unsupported claims and citation support.

2.5 Multimodal scientific retrieval

Contrastive Language–Image Pre-training (CLIP) demonstrated transferable image–text alignment at scale [17], while Bootstrapping Language–Image Pre-training (BLIP) improved both understanding and generation with filtered caption data [18]. The Scientific Multi-Modal Information Retrieval (SciMMIR) benchmark nevertheless identifies a domain gap: scientific captions tend to describe results or principles rather than the activities and scenery common in generic images [19]. For this reason, the first implementation uses Gemini Vision to produce question-aware observations and retrieval terms. These terms extend the audited text retriever instead of forming a separate vision index.

The multimodal path separates visual observations from literature evidence. Pixel intensity is not treated as calibrated temperature without radiometric metadata. The source articles carry CC BY licences, but figure-level credits are checked separately; source images are not redistributed. Questions use captions and surrounding text rather than diagnosis. This is exploratory literature retrieval, not medical-device or clinical-performance evaluation.

MR thermometry supports real-time monitoring during focused-ultrasound procedures, but performance depends on acquisition, tissue composition and motion [20, 21]. Ultrasound backscattered-energy thermometry has measured liver heating through the ablative range, but remains tissue- and method-dependent [22]. Infrared thermography is non-contact and maps surface temperature; standardisation and environmental control remain important [23, 24, 25]. The assistant must therefore separate visible features, experimental measurements and clinically established conclusions.

3 Methodology and System Design

3.1 Research design and frozen system

Figure 1 summarises the frozen architecture evaluated in this study.

The study used artefact development followed by controlled comparative evaluation. The application was frozen at commit `e416772`, tag `image-qa-v1-2026-06-23`; public FastAPI interfaces and tested parameters were not changed in response to results. Every experiment created an immutable timestamped directory with commit, seed, models, parameters, per-case output, summary and latency.

The system comprises a FastAPI backend, Vite/React client, SQLite persistence and modular ingestion, retrieval, generation and image services. The interface displays research chat, evidence cards, DOI links and model status, but evaluation concerns the underlying evidence pipeline rather than UI appearance. Corpus files and embeddings remain local; query-time Gemini calls transmit only the selected evidence excerpts or the evaluation image required for the requested operation. The prototype is therefore locally managed but not fully on-premises.

Figure 2 shows the final demonstration interface and its evidence cards.

3.2 Corpus construction and technical audit

Table 1 summarises the corpus construction and parsing audit.

The source snapshot was collected on 28 May 2026 through OpenAlex, Europe PMC, arXiv and lawful

Table 1: Corpus construction and parsing audit.

Component	Count / coverage	Audit note
Documents	272	All PDFs present
Page records	3,904	Page-level provenance
Chunks	4,243	Production 650/90 configuration
DOI metadata	82.94%	Not all sources assign a DOI
Unknown section	5.28%	Parser limitation
Visual sample	30 documents / 524 pages	5 approved; 21 minor; 4 rebuild

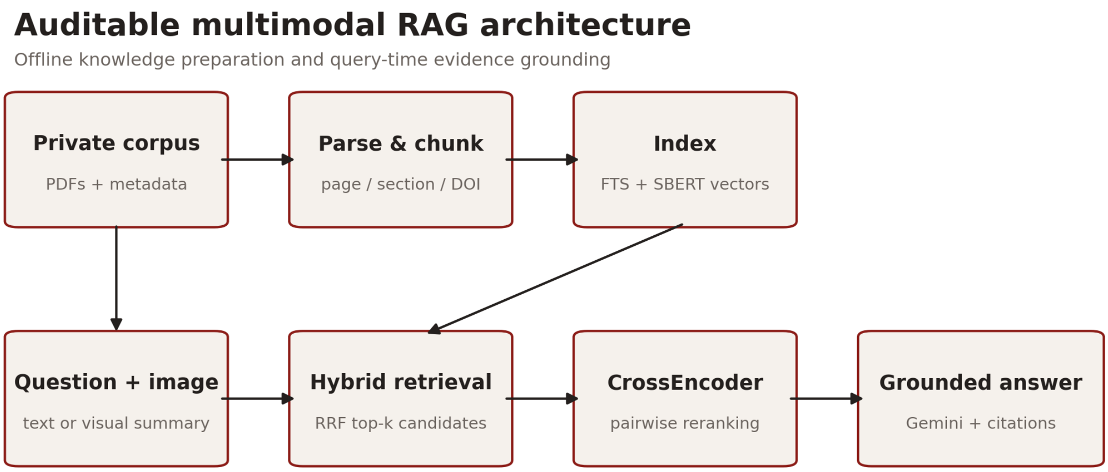

Figure 1: Auditable multimodal RAG architecture evaluated in this study.

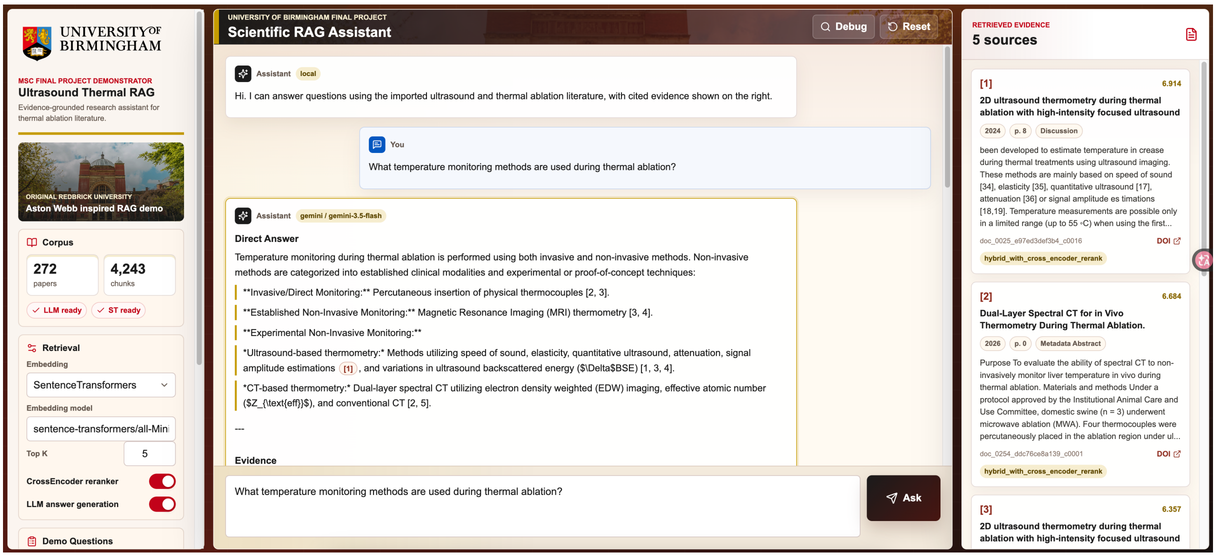

Figure 2: Demonstration interface with grounded answer and retrieved evidence cards.

publisher/repository links. Candidates required a readable, relevant PDF; duplicate DOI, inherited-key and PDF-hash records were consolidated. The build contained 272 papers from 2004–2026, including 13 obtained through authorised institutional access and kept local.

Records retained source, access and extraction metadata. PyMuPDF extracted text, coordinates and images; chunks preserved page, section and stable IDs. Corpus chunks used 650-word maximum, 120-word minimum and 90-word overlap. The separate 900/150-character upload splitter was not evaluated.

The audit checked file presence, extraction, metadata, empty text, replacement characters, short chunks and `Unknown` sections. Boundary lines repeated at least three times were flagged as possible headers/footers, not declared errors. For the deterministic sample stratified by age and length, all 524 physical pages from 30 PDFs were rendered and inspected; flagged pages were re-rendered at high resolution. Stored records were checked for page coverage, reading order, section metadata, table/equation linearisation and false reference boundaries. This author-led technical audit used automated flags and Codex to cross-check anomalies; it was not clinical or independent expert review.

3.3 Retrieval configurations

All retrieval experiments used 50 manually reviewed silver questions, subsequently cross-checked with Codex, in a fixed order with seed 3035412. The author checked each label against its source passage and online publication record before a Codex consistency check. Thirty-two were approved and 18 corrected by narrowing scope, completing an answer or replacing weak evidence; none was deleted. Crossref or repository records confirmed bibliographic identity. Without medical-specialist review, this improves traceability but not expert validity.

Full-text search in SQLite FTS supplied the lexical/BM25-style baseline. Dense retrieval used `sentence-transformers/all-MiniLM-L6-v2`; 384-dimensional corpus embeddings were precomputed and cosine similarity was evaluated locally. The model is compact enough for a laptop and requires no inference API key.

Two fusion methods were compared. RRF combined lexical and dense ranks using Equation (1):

$$\mathrm{RRF}(d) = \sum_{r \in \{\mathrm{lex},\mathrm{dense}\}} \frac{1}{60 + \mathrm{rank}_r(d)}. \tag{1}$$

Weighted fusion was also tested at lexical weights 0.50 and 0.75. Candidate pools contained up to 80 passages. CrossEncoder experiments used `cross-encoder/ms-marco-MiniLM-L-6-v2`, maximum sequence length 384, batches of 16 and the first 900 characters per candidate. Lexical, dense and RRF pools were reranked separately. Secondary sensitivity tests compared reranker pools of 10, 20, 30 and 50, and compared dense/hybrid indexes containing text only, title plus text, or title plus section plus text.

Metrics were Recall@5, @10 and @20, mean reciprocal rank (MRR), normalised discounted cumulative gain (nDCG) at 5 and 10, and end-to-end query latency. A hit required the document ID, normalised DOI or labelled chunk to match a silver source. MRR averaged $1/r_i$ for the first relevant rank r_i, with zero for a miss; nDCG discounted relevant evidence by rank. Per-question ranks were retained for paired error analysis.

3.4 Chunk ablation

Four word-based chunk configurations were reconstructed from the same page records: 350 words with 50-word overlap, the production 650/90 configuration, 650/0 and 900/150. Section and page boundaries were retained where possible. For comparability, every condition used the same questions, SentenceTransformer model, lexical index construction, RRF algorithm, candidate pool and evaluation metrics. The number and mean length of generated chunks were reported because apparent retrieval improvements can require a much larger index.

3.5 Generation and citation evaluation

Thirty answerable reviewed-silver questions were selected proportionally by type with a fixed seed; ten designed evidence-insufficient questions tested abstention. Three conditions were compared.

Condition A (no retrieval): Gemini without retrieved evidence.

Condition B (dense RAG): dense-vector retrieval with the top five passages.

Condition C (hybrid grounded RAG): RRF hybrid retrieval, CrossEncoder reranking and a grounded prompt with the top five passages.

The grounded prompt required direct answer, evidence, limitations, `Sources used`, a `[n]` marker for literature claims and abstention when evidence was insufficient. Because no sources were supplied to the no-retrieval baseline, citation precision was not applicable to that condition.

Programmatic checks validated citation indices against supplied evidence. A fixed-schema Gemini judge scored answerable-case correctness, faithfulness and completeness (0–2), and counted claims, unsupported claims, citation support and abstention. Two temperature-zero judgements were retained for each answer. Unsupported-claim rate used all cases; citation precision used answerable grounded cases; abstention accuracy used only evidence-insufficient cases. Model judgement is not expert truth.

3.6 Exploratory image question answering

From seven CC BY 4.0 corpus papers, a technical screen selected 12 complete-caption figures: four ultrasound/thermometry, four thermal-distribution and four schematic/quantitative plots. The manifest recorded source, page, figure, caption and expected document. The author checked source pages, captions, credit lines and licences online before using Codex to cross-check the records. Ten figures had no special credit; one PLOS composite used prior-study photographs, and one BioRender schematic required separate confirmation. All 12 figures were restricted to internal evaluation and excluded from the report and public package; the two flagged figures were additionally marked unsuitable for public reproduction without figure-specific confirmation. No identifying or diagnostic task was included.

Three conditions used the same question for each image.

Condition A (text only): the text question without image-derived input.

Condition B (basic metadata): the question plus locally computed dimensions, format and uncalibrated pixel statistics.

Condition C (vision summary): the question plus a Gemini Vision summary constrained to observable features, uncertainty and retrieval terms.

Each resulting query entered the same hybrid/CrossEncoder pipeline. Source Recall@5 tested paper retrieval; caption coverage, citation quality, structure, unsupported visual inference and contradiction assessed answers. Stage latency was stored separately. Captions can omit visible details and are not independent clinical labels.

3.7 Software verification and analysis

Evaluation used an Apple M1 MacBook Air (8 GB), macOS 15.2 and Python 3.10.6; the final verification suite passed all 27 tests. Selection used seed 3035412 and resampling seed 20260723. Ninety-five per cent bootstrap intervals used 10,000 resamples; paired two-sided randomisation tests used 50,000 sign swaps at $\alpha = 0.05$. The p-values were unadjusted across multiple exploratory comparisons: *nominally detectable* means unadjusted $p < 0.05$. Interpretation therefore prioritises effect sizes and confidence intervals. These tests quantify case-sampling uncertainty, not label error or external generalisation.

4 Results

4.1 Corpus and parsing quality

The automated audit found 272 PDFs, 3,904 page records and 4,243 chunks. DOI coverage was 82.94%; provenance fields were complete. There were 224 `Unknown`-section chunks (5.28%), 16 below 120 words, no replacement characters and 339 possible repeated boundary lines. Chunk length was 48–829 words (mean 522.9).

The screen classified 5 documents as approved, 21 with minor/local issues and 4 for rebuild. Two false reference boundaries omitted body text; two papers had unreliable formula/table extraction. Fifteen documents had a retained multi-section page and six had formula/table limitations. Field completeness did not establish semantic correctness; this sample identifies failures, not a corpus-wide rate.

4.2 Retrieval and reranking

Figure 3 compares retrieval and reranking performance.

Lexical MRR and Recall@5 were 0.489 and 0.56; dense retrieval reached 0.433 and 0.52. RRF was the strongest mode before reranking: MRR 0.523 (95% CI 0.401–0.649), Recall@5 0.62 (0.480–0.741) and Recall@20 0.74. Relative to dense retrieval, RRF had nominally higher MRR by 0.090 (unadjusted $p = 0.000060$) and nDCG@10 by 0.084 (unadjusted $p = 0.000140$); its Recall@5 difference was not nominally detectable. No RRF–lexical difference was nominally detectable, so the result does not establish universal hybrid superiority.

Lexical reranking raised MRR to 0.605 (unadjusted $p = 0.031959$) and nDCG@10 to 0.626 (unadjusted $p = 0.034299$); both changes were nominally detectable. RRF reranking gave MRR 0.584 and nDCG@10 0.617, but neither change was nominally detectable. Reranking therefore reordered supplied candidates but could not recover first-stage omissions.

Mean latency was 25.3 ms for lexical, 36.4 ms for dense, 62.0 ms for RRF and 321–341 ms with CrossEncoder. A top-20 reranker pool gave MRR 0.597 at 266 ms; no pool-size difference was nominally detectable. Both text-only and title-only hybrid retrieval achieved Recall@5 of 0.64, compared with 0.62 for title plus section, and no metadata difference was nominally detectable.

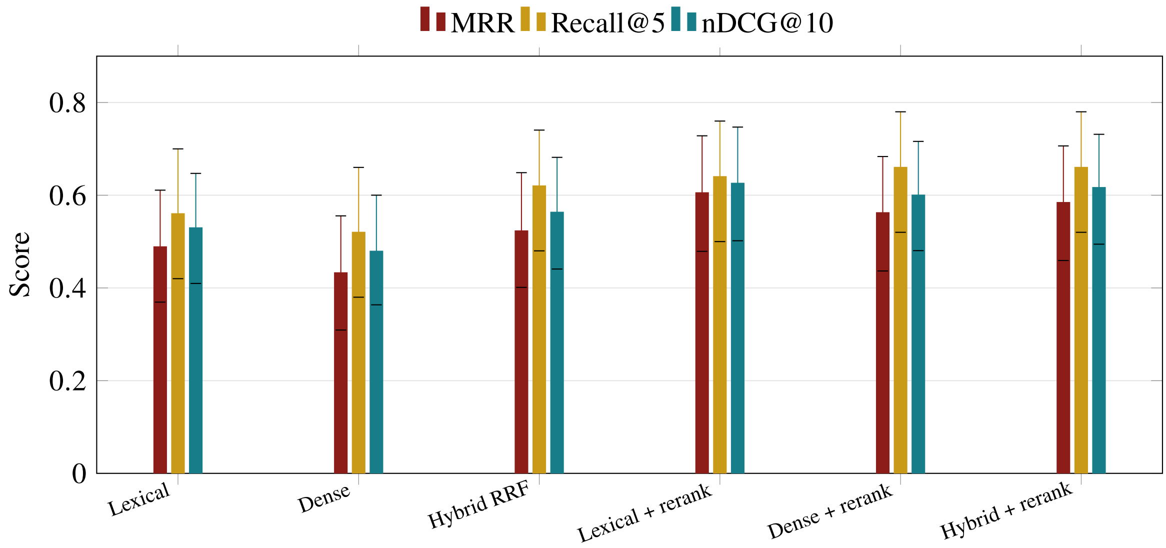

Figure 3: Retrieval and reranking performance by configuration; error bars show 95% bootstrap confidence intervals.

Table 2: Automated retrieval screening over 50 reviewed-silver questions.

Pattern	**n**	**Observation**	**Next action**
Candidate omission	13	Outside top 30	Rewrite query
Reranker demotion	2	Top-five hit moved down	Train reranker
Context cutoff	2	Ranked 6–30	Expand context
Later promotion	6	Fusion/reranker rescue	Retain stages
Single-stage hit	7	Lexical or dense only	Retain hybrid
Shared success	20	All core stages hit	None

Table 2 converts per-question ranks into an automated screening taxonomy. It identifies where to inspect, rather than proving a root cause.

4.3 Chunk ablation

Figure 4 shows the early-recall and index-size trade-off.

The 350/50 condition created 6,997 chunks and achieved the best MRR (0.574), Recall@5 (0.74) and nDCG@10 (0.618). Production 650/90 created 4,243 chunks and reached 0.523, 0.62 and 0.563. The 350/50 differences from production were nominally detectable for MRR (unadjusted $p = 0.043599$), Recall@5 (unadjusted $p = 0.031219$) and nDCG@10 (unadjusted $p = 0.019900$). Removing overlap produced 3,745 chunks, Recall@5 0.62 and Recall@20 0.78; 900/150 created 3,336 chunks and Recall@5 0.58. Thus 350/50 improved early recall by 12 points but enlarged the index by 64.9%. Production remains a resource compromise rather than the retrieval optimum observed in this sample.

4.4 Generation and citation

Generation, faithfulness and abstention results are summarised in Table 3 and Figure 5.

The no-retrieval baseline achieved mean correctness 1.467/2, faithfulness 1.633 and completeness 1.400

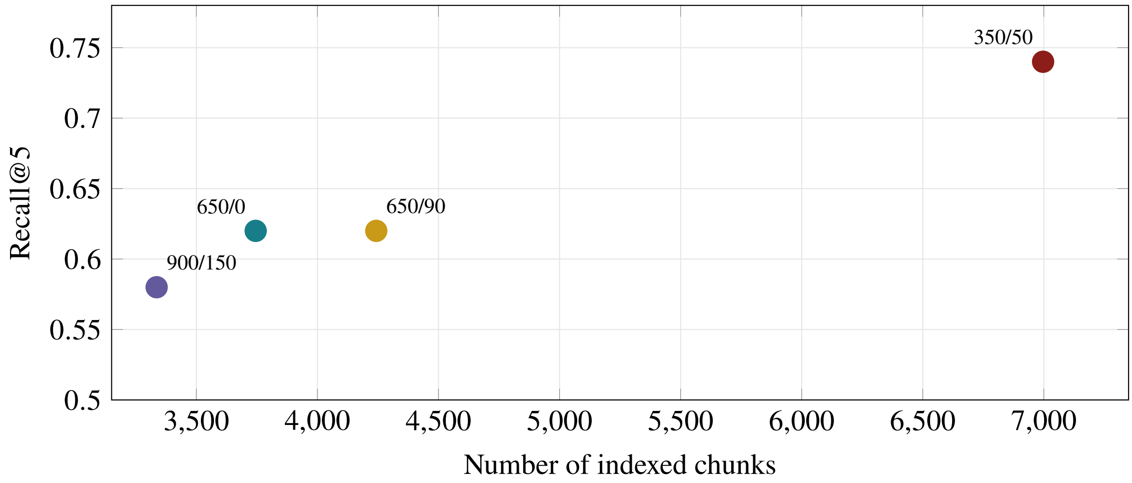

Figure 4: Recall@5 and index-size trade-off across chunk configurations.

Table 3: Generation quality on 30 answerable cases, unsupported-claim rate on all 40 cases, and abstention on ten evidence-insufficient cases.

Mode	Correct.	Faithful.	Complete.	Unsupported	Abstention
LLM only	1.467	1.633	1.400	0.269	0.700
Dense RAG	1.767	1.867	1.833	0.020	1.000
Hybrid reranked RAG	1.867	1.867	1.933	0.020	1.000

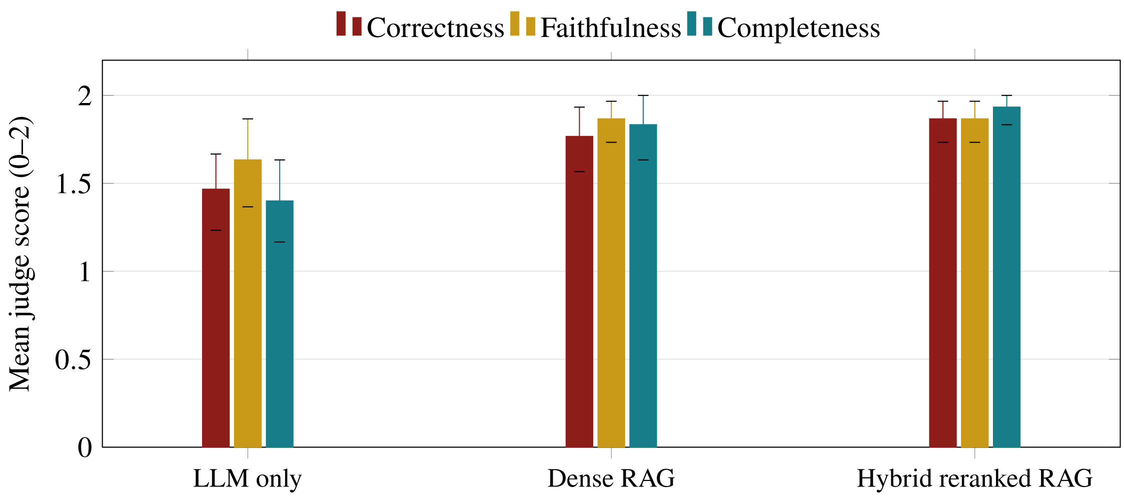

Figure 5: Mean generation scores under the three evidence conditions; error bars show 95% bootstrap confidence intervals.

Table 4: Exploratory image-QA results on 12 figures from seven CC BY 4.0 articles.

Mode	Source R@5	Caption	Citation	Structure	Contradictions
Text only	0.917	1.583	1.833	1.917	0
Basic metadata	0.917	1.750	1.833	2.000	0
Vision summary	0.917	1.750	1.917	2.000	0

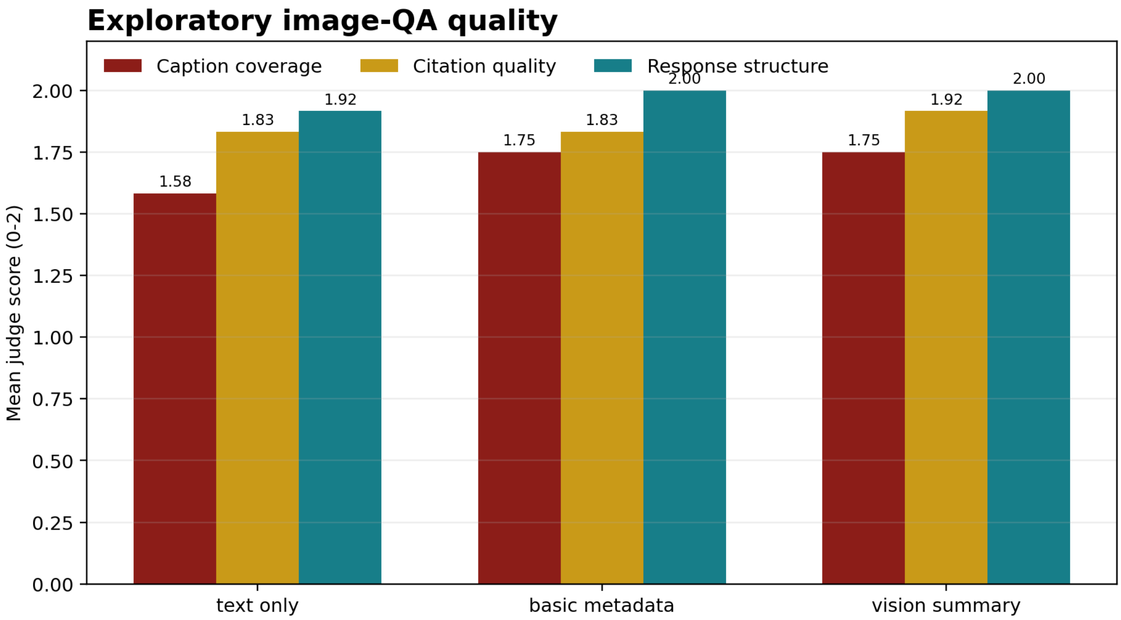

Figure 6: Image-QA quality under text, metadata and vision-summary conditions; error bars show 95% bootstrap confidence intervals.

on answerable questions. Its unsupported-claim rate was 26.9% across all cases (95% CI 10.9–43.8%), and it abstained on 70% of ten evidence-insufficient questions. No out-of-range citation marker was detected.

Dense RAG scored 1.767 for correctness, 1.867 for faithfulness and 1.833 for completeness; unsupported claims fell to 2.0% (0.5–4.0%), citation precision was 98.2% (96.0–100%) and correct abstention on the ten evidence-insufficient cases was 100%. Hybrid reranked RAG scored 1.867, 1.867 and 1.933; unsupported claims were 2.0% (0.5–4.1%), citation precision 97.6% (95.2–99.4%) and correct abstention was 100%. Relative to no retrieval, dense RAG had nominally higher completeness (unadjusted $p = 0.021560$); hybrid RAG had nominally higher correctness (unadjusted $p = 0.010020$) and completeness (unadjusted $p = 0.000120$). Faithfulness differences and all hybrid–dense differences were not nominally detectable.

The answer model was `gemini-3.5-flash-lite`. All 120 answers received two temperature-zero structured judgements from the fixed `gemini-3.6-flash` judge, producing 240 judgements. Exact repeat agreement and quadratic weighted kappa were 1.0 for every mode and metric, demonstrating deterministic reproducibility rather than independent validity. Amortised batched generation latency was approximately 0.95–1.00 s per answer and is not equivalent to single-user UI latency.

4.5 Exploratory image QA

Table 4 and Figure 6 report the exploratory image-QA comparison.

Source Recall@5 was 0.917 (95% CI 0.75–1.00) in all conditions. Text-only caption coverage was 1.583/2 (1.167–1.917); vision summary reached 1.750 (1.333–2.000) and citation quality 1.917 (1.750–2.000). No vision comparison was nominally detectable, consistent with 12 cases and caption-derived queries.

The judge counted three unsupported visual inferences for text-only answers, one with basic metadata and none with the vision summary; no contradiction or invalid citation index occurred. The zero count is a sample result, not proof of zero risk. Mean retrieval latency ranged from 543 to 635 ms, answer latency from 655 to 658 ms, and vision summarisation added 5.41 s per image.

5 Discussion

5.1 Interpretation of retrieval results

RRF increased Recall@5 by 0.06 over lexical and 0.10 over dense search, supporting the literature distinction between exact term matching and semantic representation [5, 6, 7]. Its MRR and nDCG@10 gains over dense retrieval were nominally detectable with very small unadjusted p-values, whereas no RRF–lexical difference was detected. The result supports rank fusion for this corpus [8], not a universal claim that hybrid retrieval is superior.

CrossEncoder reranking produced nominal lexical-pool gains in MRR and nDCG@10 while Recall@20 changed little, consistent with reordering within a fixed candidate pool. Candidate omission remained the largest failure category. Top-20 had a slightly higher point estimate and lower latency than top-30, but no detectable difference.

The 350/50 index produced nominal gains in MRR, Recall@5 and nDCG@10 but expanded storage by 65%. The literature identifies two related risks: splitting can remove document context, while longer input does not guarantee evidence use [12, 13]. Metadata did not uniformly improve dense retrieval, so future work should test section weighting and neighbouring-chunk expansion rather than assume prefixes are beneficial.

5.2 Generation, citation and hallucination

Grounding reduced unsupported claims and improved abstention. Its nominal gains concerned correctness or completeness, not faithfulness, and hybrid reranking did not outperform dense RAG. This supports component-level evaluation advocated by RAGAS and ARES [4, 15]. Reviewed-silver labels and a Gemini-family judge still prevent claims of scientific correctness.

External evidence does not eliminate hallucination [2, 3, 16]. A generator can attach a valid index to the wrong claim or overgeneralise an experiment. Code verifies citation syntax and range, not semantic entailment; independent sentence-level review remains necessary.

5.3 Multimodal interpretation

The image experiment did not support the hypothesis that a visual summary would increase source Recall@5. Caption-derived questions created a ceiling effect, consistent with the gap between generic image–language tasks [17, 18] and scientific-figure retrieval [19]. No quality difference was nominally detectable; the feature's present value is explicit observation and uncertainty, not higher retrieval accuracy, at 5.4 s additional latency.

Separating observations from literature claims is useful, but summary errors can influence retrieval and generation. Captions are not independent labels because they guided the questions. Future work should add independent questions, panel crops, axis/legend OCR and scientific image-text embeddings.

5.4 Threats to validity

The 50 questions are manually reviewed silver labels subsequently cross-checked with Codex, not exhaustive relevance judgements or expert gold labels. The author checked online primary sources and publication records before this secondary check. Without medical expertise, this cannot establish clinical validity or replace qualified expert judgement. A valid unlisted paper may still be counted wrong, while source-level matching can reward a document whose retrieved chunk lacks the required sentence.

Thirty answerable cases and 12 images support comparison, not general biomedical claims. Designed unanswerable questions may make abstention easier. Gemini-family judging introduces rubric and family bias; repeated temperature-zero outputs measure reproducibility, not independent human agreement. The unadjusted exploratory tests cover multiple metrics and configurations, increasing false-positive risk for marginal p-values; effect sizes and confidence intervals are therefore more informative than a binary threshold.

Audit metrics are proxies: complete fields need not be correct, and repeated boundary lines need not be harmful. The author-led 524-page screen, supported by automated flags and Codex cross-checking, exposed false reference stops, coarse page-level sections and damaged table/equation structure, but a 30-document purposive sample cannot establish the error rate of the full corpus. Scientific PDFs remain vulnerable to publisher-specific layout errors [9, 10]. The rights check cannot replace independent confirmation of third-party panels, and institution-accessible material requires authorised use.

Latency was measured on one Apple computer after loading, not under multi-user load. API latency and quotas remain provider- and network-dependent.

5.5 Objective traceability

Table 5 shows how each objective is evidenced rather than inferred from the interface.

Table 5: Objective traceability.

Objective	**Method / result evidence**	**Conclusion**
O1 corpus	ID/metadata audit	272-paper traceable corpus
O2 parsing	30 PDFs / 524-page screen	26 usable; 4 require rebuild
O3 chunking	Four-condition ablation	650/90 trades size for recall
O4 retrieval	Retrieval/reranking tests	Fusion broadens; reranking reorders
O5 grounding	40-question test	Unsupported claims reduced
O6 multimodal	12-figure comparison	Exploratory, not validated

6 Impact Statement

The immediate benefit is a more auditable way for students and researchers to explore a specialised scientific collection. Instead of accepting an uncited answer from a general model, a user can inspect the retrieved passage, paper title, DOI, page and chunk identifier. This can reduce the time spent locating initial evidence and makes disagreements easier to investigate. Institution-accessible papers can also extend the knowledge base beyond publicly indexed full text without incorporating their contents into model parameters.

The primary stakeholders are postgraduate researchers, supervisors and research groups working with ultrasound or thermal-imaging literature. Librarians and institutional IT teams are relevant to lawful access, retention and deployment. Authors and publishers are affected by how full text and figures are stored and displayed. Patients are not direct users of this prototype, but could be indirectly affected if unvalidated research summaries were mistaken for clinical advice.

Implementation should therefore remain research-oriented. A university deployment would require authenticated access, encrypted storage, document-level licence rules, audit logging, deletion processes and periodic corpus review. Its benefits should be measured through evidence-finding time, retrieval coverage and citation support, not through unsupported claims of better clinical decisions. Local embeddings reduce repeated transmission of complete documents, but selected excerpts and evaluation images are still sent to Gemini. API prompts should contain only the minimum required evidence, and the prototype should not be described as fully on-premises.

The likely economic benefit is researcher time saved when locating and checking papers. The design could also support controlled institutional or industrial knowledge management, but this prototype is neither commercial nor clinical. Deployment would add storage, API, maintenance and rights-management costs. Environmental effects were not measured: local inference may reduce network transfer, while model computation and document storage still consume energy.

The major risks are copyright misuse, confidential-data ingestion, automation bias, biased or incomplete literature, false citations and visual over-interpretation. Controls include authorised corpus acquisition, CC BY images for evaluation, no patient identifiers, explicit uncertainty, evidence cards, citation validation and a prohibition on diagnosis or treatment recommendations. The system can support literature navigation, but accountability for scientific interpretation remains with the researcher.

7 Conclusions

This project produced an auditable multimodal RAG demonstrator over 272 papers and 4,243 provenance-preserving chunks. For RQ1, no configuration dominated every criterion. RRF offered the strongest pre-reranking trade-off at 62 ms (Recall@5 0.62), while lexical-pool CrossEncoder reranking achieved the highest MRR, 0.605, at 341 ms. RRF's MRR and nDCG@10 advantages over dense retrieval were nominally detectable, but its differences from lexical retrieval were not.

For RQ2, 350/50 produced the highest Recall@5, 0.74, but 65% more chunks than production; its MRR, Recall@5 and nDCG@10 gains were nominally detectable. The selected 650/90 configuration is therefore a resource compromise rather than an optimum.

For RQ3, grounding reduced unsupported claims from 26.9% to approximately 2.0%, correctly abstained on all ten evidence-insufficient cases and achieved 97.6–98.2% citation precision. Hybrid RAG produced nominal correctness and completeness gains, but no faithfulness or hybrid–dense difference. For RQ4, vision summaries did not improve Recall@5 beyond 0.917 or produce a detectable quality difference. The feature remains exploratory. Future work should prioritise corpus, retrieval and evidence quality; expert labels and figure-specific rights checks remain necessary.

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A Abbreviations and Symbols

API: application programming interface.

BM25: Best Matching 25 probabilistic lexical ranking function.

BLIP: Bootstrapping Language–Image Pre-training.

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CI: confidence interval.

CLIP: Contrastive Language–Image Pre-training.

DOI: digital object identifier.

FTS: full-text search.

HIFU: high-intensity focused ultrasound.

IRT: infrared thermography.

LLM: large language model.

MR/MRI: magnetic resonance/magnetic resonance imaging.

MRR: mean reciprocal rank.

nDCG: normalised discounted cumulative gain.

PDF: Portable Document Format.

PRFS: proton resonance frequency shift.

QA: question answering.

RAG: retrieval-augmented generation.

RQ: research question.

RRF: reciprocal rank fusion.

SBERT: Sentence-BERT.

SciMMIR: Scientific Multi-Modal Information Retrieval.

TA: thermoacoustic.

UST: ultrasound tomography.

Recall@k: proportion of questions with labelled evidence in the first k results.

top-k: the first k ranked retrieval results.

B Reproducibility and Review Boundary

The evaluated software state is Git commit `e416772`, tag `image-qa-v1-2026-06-23`. Runs record models, parameters, seeds, input hashes, per-case output, summaries and latency. Privacy-safe source code and reproducibility materials are available at `https://github.com/ChrisWu11/multimodal-rag`; private PDFs, evidence text, source figures, embeddings, the database, local paths and API credentials are excluded.

Completed checks include the corpus audit, confidence intervals, paired tests, citation-index validation, verification of all 25 references, article-level licence checks and source-page rights checks for 12 figures. Bibliographic identity does not prove claim support, and an article licence does not prove that every panel is original.

The 50-label review recorded 32 approvals and 18 corrections; expert sign-off remains separate. The PDF audit found 5 approved, 21 minor, and 4 rebuild cases. All source figures were excluded from public release. The module permits the declared use of Codex and Gemini, but wording remains the author's responsibility. No patient-identifying data are processed, and the system is not a clinical decision tool.